

Zadanie 30

Oblicz granicę $\lim_{n \rightarrow \infty} \left(\sqrt[n]{\left(\frac{2}{5}\right)^n + (0,6)^n} + \sqrt[n]{2^{3n+5}} \right)$.

① Korzystamy z twierdzenia o trzech ciągach.

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2}{5}\right)^n + (0,6)^n} :$$

$$\sqrt[n]{\left(\frac{3}{5}\right)^n} \leq \sqrt[n]{\left(\frac{2}{5}\right)^n + \left(\frac{3}{5}\right)^n} \leq \sqrt[n]{\left(\frac{3}{5}\right)^n + \left(\frac{3}{5}\right)^n}$$

$$a_n = \sqrt[n]{\left(\frac{3}{5}\right)^n} \quad \lim_{n \rightarrow \infty} a_n = \frac{3}{5}$$

$$b_n = \sqrt[n]{\left(\frac{2}{5}\right)^n + \left(\frac{3}{5}\right)^n} \quad \Rightarrow \quad \lim_{n \rightarrow \infty} b_n = \frac{3}{5}$$

$$c_n = \sqrt[n]{\left(\frac{3}{5}\right)^n + \left(\frac{3}{5}\right)^n} \quad \lim_{n \rightarrow \infty} c_n = \frac{3}{5}$$

② Obliczamy $\lim_{n \rightarrow \infty} \sqrt[n]{2^{3n+5}}$,

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^{3n+5}} = \lim_{n \rightarrow \infty} \sqrt[n]{8^n \cdot 32} = \lim_{n \rightarrow \infty} 8 \cdot \sqrt[n]{32} = 8 \cdot 1 = 8$$

③ Obliczamy granicę.

$$\lim_{n \rightarrow \infty} \left(\sqrt[n]{\left(\frac{2}{5}\right)^n + (0,6)^n} + \sqrt[n]{2^{3n+5}} \right) = \frac{3}{5} + 8 = 8\frac{3}{5}$$

$$\text{Odp.: } \lim_{n \rightarrow \infty} \left(\sqrt[n]{\left(\frac{2}{5}\right)^n + (0,6)^n} + \sqrt[n]{2^{3n+5}} \right) = 8\frac{3}{5}.$$